# Question_4:

## 1. Verilog code :

- **Module code:**

```verilog
module spr(clk,rst,din,addr,wr_en,rd_en,blk_select,addr_en,dout_en,dout,parity_out);
parameter MEM_WIDTH=16;
parameter MEM_DEPTH=1024;
parameter ADDR_SIZE=10;
parameter ADDR_PIPELINE ="FALSE";
parameter DOUT_PIPELINE="TRUE";
parameter PARITY_ENABLE=1;
////////////////////////////
input clk,rst;
input rd_en,wr_en,blk_select,addr_en,dout_en;
input [MEM_WIDTH-1:0] din;
input [ADDR_SIZE-1:0] addr;
output  reg parity_out;
output   [MEM_WIDTH-1:0] dout;
//////////////////////////////
reg [MEM_WIDTH-1:0] mem[MEM_DEPTH-1:0] ;
//////////////////////////////
reg [ADDR_SIZE-1:0] addr_ff;
wire [ADDR_SIZE-1:0] addr_mem;
reg [MEM_WIDTH-1:0] dout_ff;
reg [MEM_WIDTH-1:0] dout_mem;
////////////////////////////////
assign addr_mem=(ADDR_PIPELINE=="FALSE")?addr:addr_ff;
assign dout=(DOUT_PIPELINE=="FALSE")?dout_mem:dout_ff;
always @(posedge clk) begin
if(rst)
addr_ff<=0;
 else if((addr_en))
begin
addr_ff<=addr;
end
end

always @(posedge clk ) begin
if(rst)
dout_ff<=0;
else if (dout_en)begin
dout_ff<=dout_mem;
end
end
always @(posedge clk) begin
    if(rst) begin
            dout_mem <= 0;
parity_out<=0;
        end
else begin
            if (blk_select) begin
                if (wr_en) begin
                    mem[addr_mem] <= din;
                end
                if (rd_en) begin
                    dout_mem <= mem[addr_mem];
                end

if(PARITY_ENABLE==1) begin
parity_out<=^mem[addr_mem];
end
         end
        end
end
endmodule
```

- **Test bench:**

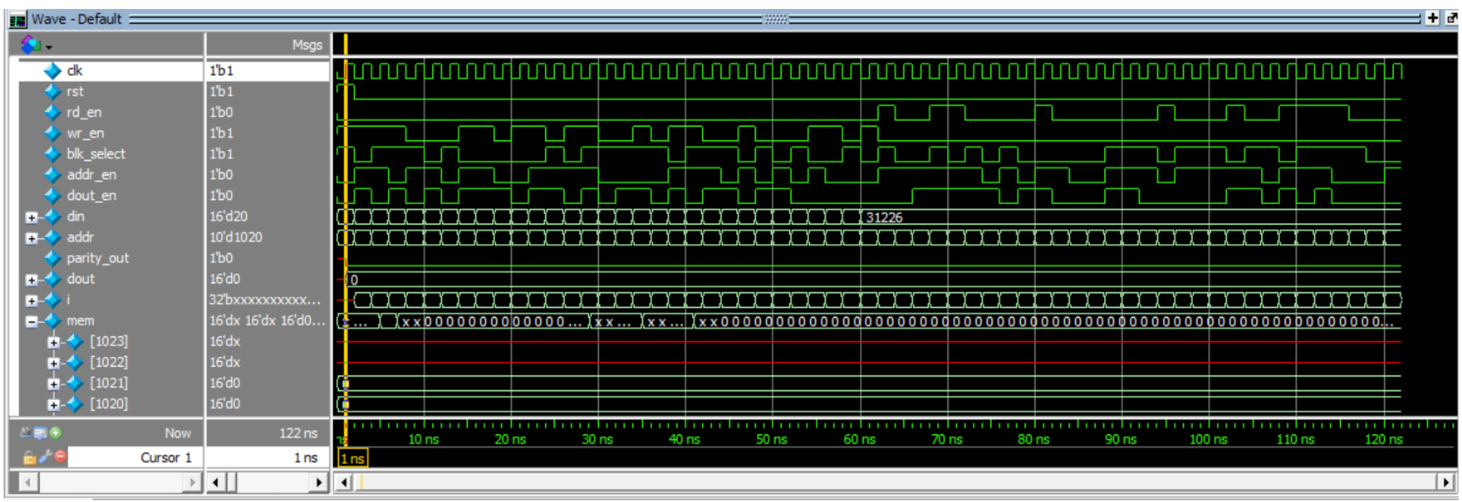
```
module spr_tb();
parameter MEM_WIDTH=16;
parameter MEM_DEPTH=1024;
parameter ADDR_SIZE=10;
reg clk,rst;
reg rd_en,wr_en,blk_select,addr_en,dout_en;
reg [MEM_WIDTH-1:0] din;
reg [ADDR_SIZE-1:0] addr;
wire parity_out;
wire [MEM_WIDTH-1:0] dout;
////////////////////
spr SPR (clk,rst,din,addr,wr_en,rd_en,blk_select,addr_en,dout_en,dout,parity_out);
//////////////////////////////////
initial begin
    clk=0;
     forever
     #1 clk=~clk;
end
integer i;
initial begin
    $readmemh("mem.dat",SPR.mem);
    rst=1;
    rd_en=0;
    wr_en=1;
    dout_en=0;
    blk_select=1;
addr=10'd1020;
din=16'd20;
addr_en=0;
    @(negedge clk);
    rst=0;

    rd_en=0;

        for(i=0;i<30;i=i+1)
    begin
        blk_select=$random;

        addr=$random;
        wr_en=$random;
addr_en=$random;
dout_en=$random;
        din=$random;
        @(negedge clk);
    end
//////////////////////////////////////////////////
    wr_en=0;


    for(i=0;i<30;i=i+1) begin
        blk_select=$random;
        addr=$random;
        rd_en=$random;
        addr_en=$random;
dout_en=$random;

        @(negedge clk);
    end

    $stop;
end
endmodule
```

- **Do file:**

```
vlib work
vlog Q4_1.v  Q4_1_tb.v
vsim -voptargs=+acc work.spr_tb
add wave *
run -all
#quit -sim
```

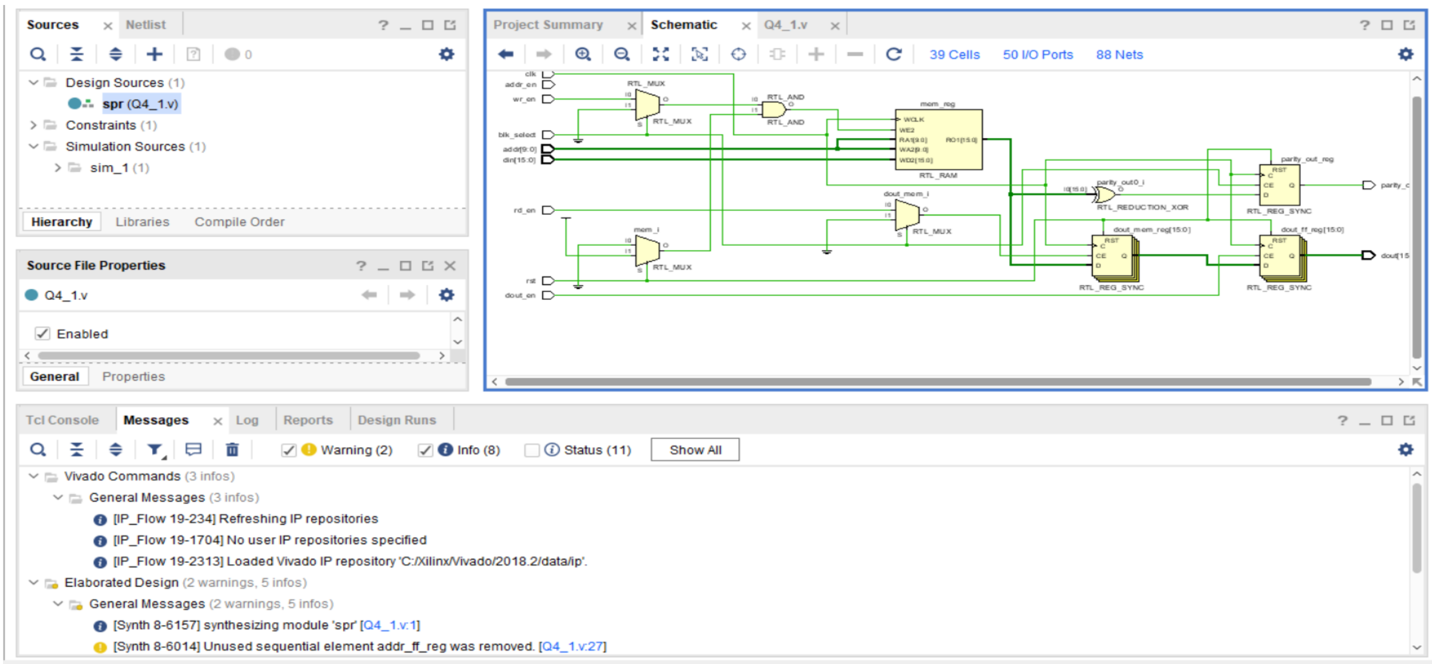
## 2. Waveform snippet:

- **Waveform:**

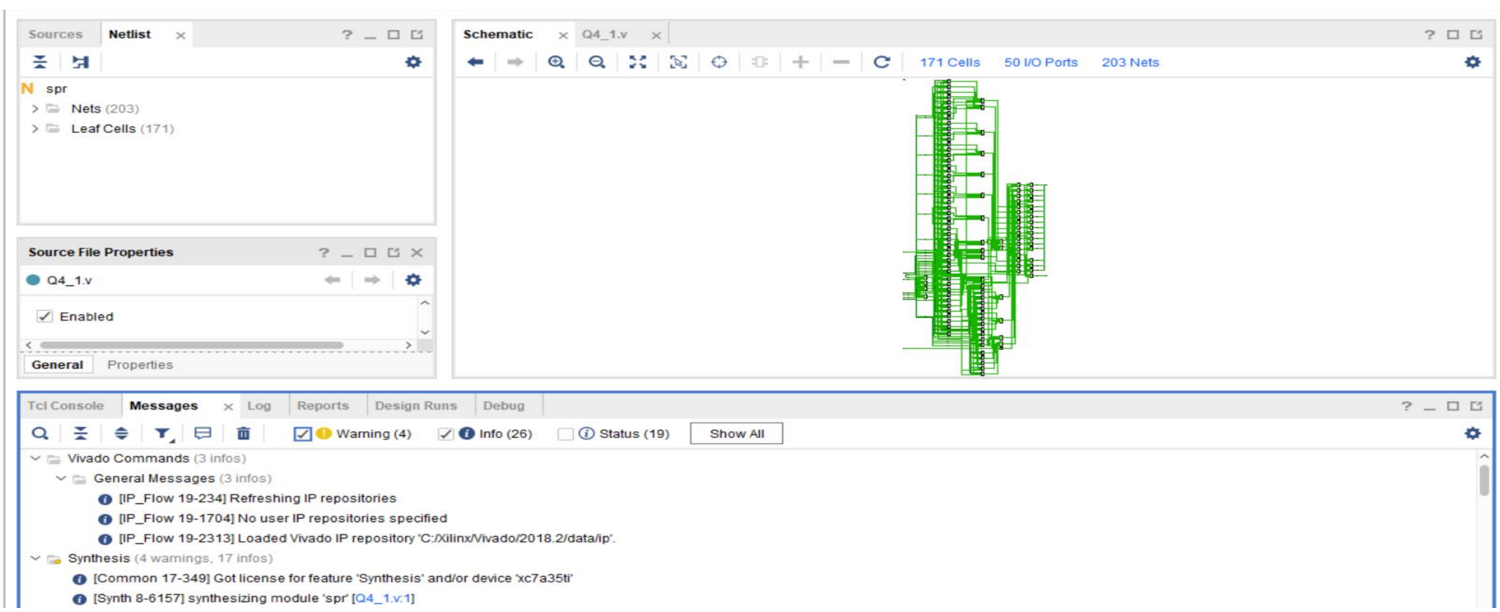



## 3. Synthesis tool:

- **Elaboration:**




- **Synthesis :**




Hierarchy

| Name | Slice LUTs (20800) | Slice Registers (41600) | F7 Muxes (16300) | F8 Muxes (8150) | Bonded IOB (106) | BUFGCTRL (32) |
|---|---|---|---|---|---|---|
| N spr | 280 | 33 | 128 | 64 | 49 | 1 |

Design Timing Summary

| Setup | | Hold | | Pulse Width | |
|---|---|---|---|---|---|
| Worst Negative Slack (WNS): | 5.335 ns | Worst Hold Slack (WHS): | 0.154 ns | Worst Pulse Width Slack (WPWS): | 3.750 ns |
| Total Negative Slack (TNS): | 0.000 ns | Total Hold Slack (THS): | 0.000 ns | Total Pulse Width Negative Slack (TPWS): | 0.000 ns |
| Number of Failing Endpoints: | 0 | Number of Failing Endpoints: | 0 | Number of Failing Endpoints: | 0 |
| Total Number of Endpoints: | 33 | Total Number of Endpoints: | 33 | Total Number of Endpoints: | 290 |

All user specified timing constraints are met.

- **Netlist:**

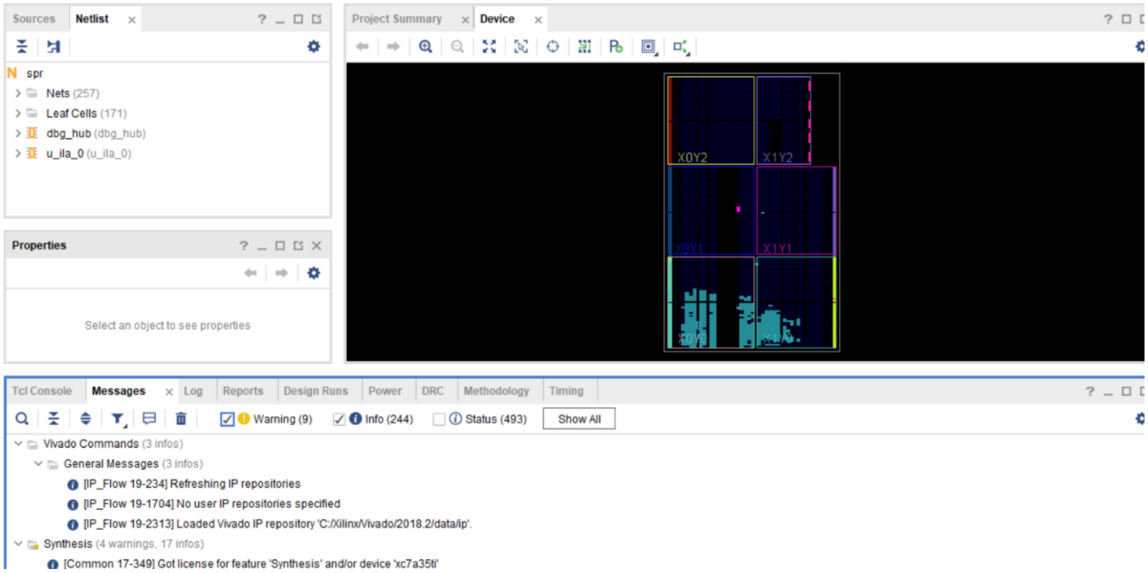
```
// ---------------------------------------------------------------------------------------------
`timescale 1 ps / 1 ps

(* ADDR_PIPELINE = "FALSE" *) (* ADDR_SIZE = "10" *) (* DOUT_PIPELINE = "TRUE" *)
(* MEM_DEPTH = "1024" *) (* MEM_WIDTH = "16" *) (* PARITY_ENABLE = "1" *)
(* STRUCTURAL_NETLIST = "yes" *)
module spr
   (clk,
    rst,
    din,
    addr,
    wr_en,
    rd_en,
    blk_select,
    addr_en,
    dout_en,
    dout,
    parity_out);
  input clk;
  input rst;
  input [15:0]din;
  input [9:0]addr;
  input wr_en;
  input rd_en;
  input blk_select;
  input addr_en;
  input dout_en;
  output [15:0]dout;
  output parity_out;

  wire [9:0]addr;
  wire [9:0]addr_IBUF;
  wire blk_select;
  wire blk_select_IBUF;
  wire clk;
  wire clk_IBUF;
  wire clk_IBUF_BUFG;
  wire [15:0]din;
  wire [15:0]din_IBUF;
  wire [15:0]dout;
  wire [15:0]dout_OBUF;
  wire dout_en;
  wire dout_en_IBUF;
  wire [15:0]dout_mem;
  wire [15:0]dout_mem0;
  wire \dout_mem[15]_i_1_n_0 ;
  wire mem_reg_0_255_0_0_i_1_n_0;
  wire mem_reg_0_255_0_0_n_0;
  wire mem_reg_0_255_10_10_n_0;
  wire mem_reg_0_255_11_11_n_0;
  wire mem_reg_0_255_12_12_n_0;
  wire mem_reg_0_255_13_13_n_0;
  wire mem_reg_0_255_14_14_n_0;
  wire mem_reg_0_255_15_15_n_0;
  wire mem_reg_0_255_1_1_n_0;
  wire mem_reg_0_255_2_2_n_0;
  wire mem_reg_0_255_3_3_n_0;
  wire mem_reg_0_255_4_4_n_0;
  wire mem_reg_0_255_5_5_n_0;
  wire mem_reg_0_255_6_6_n_0;
  wire mem_reg_0_255_7_7_n_0;
  wire mem_reg_0_255_8_8_n_0;
  wire mem_reg_0_255_9_9_n_0;
  wire mem_reg_256_511_0_0_i_1_n_0;
  wire mem_reg_256_511_0_0_n_0;
  wire mem_reg_256_511_10_10_n_0;
  wire mem_reg_256_511_11_11_n_0;
  wire mem_reg_256_511_12_12_n_0;
  wire mem_reg_256_511_13_13_n_0;
  wire mem_reg_256_511_14_14_n_0;
  wire mem_reg_256_511_15_15_n_0;
```

```
wire mem_reg_256_511_1_1_n_0;
wire mem_reg_256_511_2_2_n_0;
wire mem_reg_256_511_3_3_n_0;
wire mem_reg_256_511_4_4_n_0;
wire mem_reg_256_511_5_5_n_0;
wire mem_reg_256_511_6_6_n_0;
wire mem_reg_256_511_7_7_n_0;
wire mem_reg_256_511_8_8_n_0;
wire mem_reg_256_511_9_9_n_0;
wire mem_reg_512_767_0_0_i_1_n_0;
wire mem_reg_512_767_0_0_n_0;
wire mem_reg_512_767_10_10_n_0;
wire mem_reg_512_767_11_11_n_0;
wire mem_reg_512_767_12_12_n_0;
wire mem_reg_512_767_13_13_n_0;
wire mem_reg_512_767_14_14_n_0;
wire mem_reg_512_767_15_15_n_0;
wire mem_reg_512_767_1_1_n_0;
wire mem_reg_512_767_2_2_n_0;
wire mem_reg_512_767_3_3_n_0;
wire mem_reg_512_767_4_4_n_0;
wire mem_reg_512_767_5_5_n_0;
wire mem_reg_512_767_6_6_n_0;
wire mem_reg_512_767_7_7_n_0;
wire mem_reg_512_767_8_8_n_0;
wire mem_reg_512_767_9_9_n_0;
wire mem_reg_768_1023_0_0_i_1_n_0;
wire mem_reg_768_1023_0_0_n_0;
wire mem_reg_768_1023_10_10_n_0;
wire mem_reg_768_1023_11_11_n_0;
wire mem_reg_768_1023_12_12_n_0;
wire mem_reg_768_1023_13_13_n_0;
wire mem_reg_768_1023_14_14_n_0;
wire mem_reg_768_1023_15_15_n_0;
wire mem_reg_768_1023_1_1_n_0;
```

- **Constrains file:**

```
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe2]
set_property port_width 16 [get_debug_ports u_ila_0/probe2]
connect_debug_port u_ila_0/probe2 [get_nets [list {din_IBUF[0]} {din_IBUF[1]} {din_IBUF[
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe3]
set_property port_width 1 [get_debug_ports u_ila_0/probe3]
connect_debug_port u_ila_0/probe3 [get_nets [list blk_select_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe4]
set_property port_width 1 [get_debug_ports u_ila_0/probe4]
connect_debug_port u_ila_0/probe4 [get_nets [list clk_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe5]
set_property port_width 1 [get_debug_ports u_ila_0/probe5]
connect_debug_port u_ila_0/probe5 [get_nets [list dout_en_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe6]
set_property port_width 1 [get_debug_ports u_ila_0/probe6]
connect_debug_port u_ila_0/probe6 [get_nets [list parity_out_OBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe7]
set_property port_width 1 [get_debug_ports u_ila_0/probe7]
connect_debug_port u_ila_0/probe7 [get_nets [list rd_en_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe8]
set_property port_width 1 [get_debug_ports u_ila_0/probe8]
connect_debug_port u_ila_0/probe8 [get_nets [list rst_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe9]
set_property port_width 1 [get_debug_ports u_ila_0/probe9]
connect_debug_port u_ila_0/probe9 [get_nets [list wr_en_IBUF]]
set_property C_CLK_INPUT_FREQ_HZ 300000000 [get_debug_cores dbg_hub]
set_property C_ENABLE_CLK_DIVIDER false [get_debug_cores dbg_hub]
set_property C_USER_SCAN_CHAIN 1 [get_debug_cores dbg_hub]
```

- **Implementation:**




| Setup | | Hold | | Pulse Width | |
|---|---|---|---|---|---|
| Worst Negative Slack (WNS): | 2.663 ns | Worst Hold Slack (WHS): | 0.030 ns | Worst Pulse Width Slack (WPWS): | 3.750 ns |
| Total Negative Slack (TNS): | 0.000 ns | Total Hold Slack (THS): | 0.000 ns | Total Pulse Width Negative Slack (TPWS): | 0.000 ns |
| Number of Failing Endpoints: | 0 | Number of Failing Endpoints: | 0 | Number of Failing Endpoints: | 0 |
| Total Number of Endpoints: | 4621 | Total Number of Endpoints: | 4605 | Total Number of Endpoints: | 2956 |

All user specified timing constraints are met.

| Name | Slice LUTs (20800) | Slice Registers (41600) | F7 Muxes (16300) | F8 Muxes (8150) | Slice (8150) | LUT as Logic (20800) | LUT as Memory (9600) | LUT Flip Flop Pairs (20800) | Block RAM Tile (50) | Bonded IOB (106) |
|---|---|---|---|---|---|---|---|---|---|---|
| spr | 1787 | 2392 | 143 | 64 | 796 | 1361 | 426 | 939 | 1.5 | 49 |
| dbg_hub (dbg_hub) | 476 | 727 | 0 | 0 | 231 | 452 | 24 | 316 | 0 | 0 |
| u_ila_0 (u_ila_0) | 1031 | 1632 | 15 | 0 | 489 | 885 | 146 | 615 | 1.5 | 0 |